

Onishi Tatsuki

Data Science AI Innovation Research Promotion Center

Shiga University

1-1-1, Bamba, Hikone, Shiga, 522-8522 Japan

Telephone: +81-749-27-1023

E-mail: bougtoir@gmail.com

ORCID: 0000-0001-7261-9062

11 June 2026

The Editor

Journal of Economic Growth

Dear Professor Galor,

I am pleased to submit for your consideration the enclosed manuscript, "**What the Solow Residual Has Been Hiding: Tempo Drift and the Missing Intangible Share in National Capital Stocks**", for publication in the *Journal of Economic Growth*.

The paper addresses a fundamental question in growth accounting: how much of what we call TFP is a genuine productivity residual, and how much is a book-keeping artefact of mis-timed capital stocks and omitted intangibles? I show that two parameters — a time-varying investment-to-output lag $\mu(t)$ and an intangible-capital share β — jointly account for a measurable share of measured TFP variation across 39 OECD and middle-income economies (Penn World Table 10.01, World Bank CWON, 1995–2019). Letting μ drift reduces out-of-sample GDP forecast error by 13 %, and a Solow-residual decomposition reveals that up to 30 % of TFP-growth variance can be re-attributed to tempo drift and missing intangibles rather than genuine innovation. When both corrections are applied, production-side (flow) and wealth-side (stock) national accounts agree to within 1–2 % for most countries — the first empirical success, to my

knowledge, of the "unified national-wealth accounting" programme that Stiglitz, Sen, and Fitoussi (2009) called for.

I believe this work is well suited to the *Journal of Economic Growth* for three reasons. First, it addresses the measurement foundations of long-run growth accounting — specifically, how the Solow residual confounds genuine productivity innovation with book-keeping artefacts arising from mis-timed capital stocks and omitted intangibles. This is a topic of central importance to the growth-accounting tradition that the *Journal* has advanced since its founding. Second, the identification strategy exploits a structural analogy with the demographic tempo literature (Bongaarts & Feeney, 1998; Goldstein, Lutz, & Scherbov, 2003), generating falsifiable predictions that are confirmed by the data: the joint flow–stock loss function collapses the (μ, β) parameter space from a broad ridge to a point for 28 of 39 countries, and the resulting parameter estimates move in the directions predicted by economic theory. The tempo adjustment was itself a neglected insight within demography — dormant from the 1970s until its 1998 revival — and to the best of my knowledge this paper is the first to transplant the framework to capital accounting and GDP measurement, a domain where the quantitative stakes are orders of magnitude larger. Third, the counterfactual wealth calculations have concrete implications for the Beyond-GDP debate, showing that official statistics systematically understate the productive capital base of innovation-intensive economies by 0.3–1.1 %.

The manuscript is approximately 10,000 words of body text, with eleven figures and five tables.

I confirm the following:

- This work is original and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All data sources (Penn World Table 10.01; World Bank CWON; World Bank WDI) are publicly available; analysis code and intermediate outputs are archived in the public companion repository.
- I am the sole author.
- I have no competing interests, financial or otherwise, related to this work.

- I used generative AI to assist with formatting the text, choosing words, and writing analysis code; this is documented in the manuscript.

I would be grateful for your consideration.

Yours sincerely,

Onishi Tatsuki

Data Science AI Innovation Research Promotion Center

Shiga University

bougtoir@gmail.com